

GYAN SETU

Chapter-2, Polynomials Notes

What is a Polynomial?

A polynomial is an expression made up of variables (like x) and numbers, using only addition, subtraction, and multiplication (no division by variable, no square roots of variable).

Examples of Polynomials:

- $-2x + 3$ (Linear)
- $-x^2 - 5x + 6$ (Quadratic)
- $-2x^3 + x - 7$ (Cubic)

Not Polynomials: $1/x$, $\sqrt{x} + 2$, because powers must be whole numbers.

Terms used in Polynomials

- **Constant Polynomial:** No variable (e.g., 5)
- **Linear Polynomial:** Degree 1 (e.g., $x + 1$)
- **Quadratic Polynomial:** Degree 2 (e.g., $x^2 + 3x + 2$)
- **Cubic Polynomial:** Degree 3 (e.g., $x^3 - 2x + 4$)
- **Degree of Polynomial:** Highest power of variable.

Zeros of a Polynomial

Zero of a polynomial is the value of x which makes the polynomial equal to 0.

Example: For $p(x) = x^2 - 5x + 6$, $p(2) = 0$ and $p(3) = 0$, so zeroes are 2 and 3.

Division Algorithm for Polynomials

If we divide one polynomial $p(x)$ by another polynomial $g(x)$, then:

$$p(x) = g(x) \times q(x) + r(x)$$

where $q(x)$ is quotient and $r(x)$ is remainder. Remainder degree $< g(x)$.

Factor Theorem:

$x - a$ is a factor of the polynomial $p(x)$, if $p(a) = 0$. Also, if $x - a$ is a factor of $p(x)$, then $p(a) = 0$.

Important Algebraic Identities

(i) $(x + a)(x + b) = x^2 + (a + b)x + ab$

(ii) $(x + y)^2 = x^2 + 2xy + y^2$

(iii) $(x - y)^2 = x^2 - 2xy + y^2$

(iv) $x^2 - y^2 = (x + y)(x - y)$

(v) $(x + y)^3 = x^3 + y^3 + 3xy(x + y)$

(vi) $(x - y)^3 = x^3 - y^3 - 3xy(x - y)$

(vii) $(x + y + z)^2 = x^2 + y^2 + z^2 + 2xy + 2yz + 2zx$

(viii) $x^3 + y^3 + z^3 - 3xyz = (x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)$
